

PRODUCT REQUIREMENTS DOCUMENT

TrekTrack AI

AI-Enhanced Mobile Trek Tracking & Terrain-Aware Distance Measurement System

Version 1.0	Date March 2025	Status Draft	Platform iOS & Android
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Product Overview

1.1 Product Name

TrekTrack AI — AI-Enhanced Mobile Trek Tracking and Terrain-Aware Distance Measurement System

1.2 Problem Statement

Trekkers and hikers in mountainous and forested environments are underserved by existing navigation and fitness applications. Current solutions compute only 2D horizontal distances, misclassify trekking as vehicular movement, and cannot handle the GPS noise introduced by dense canopy and rocky terrain. The result is inaccurate trek statistics, unreliable activity logs, and a poor user experience for a growing outdoor recreation market.

1.3 Vision & Purpose

TrekTrack AI aims to become the definitive digital companion for outdoor trekkers worldwide — delivering accurate, terrain-aware distance measurement, intelligent activity classification, and rich analytics through a mobile-first experience. The product bridges the gap between generic navigation platforms and the specialized requirements of off-road trekking.

1.4 Target Users

User Segment	Description	Scale
Recreational Trekkers	Weekend hikers and nature enthusiasts tracking personal fitness progress	Primary
Adventure Athletes	Trail runners, mountaineers seeking precise performance metrics	Primary
Guided Tour Groups	Professional guides managing multi-member trek sessions	Secondary
Conservation Researchers	Field scientists logging routes in protected natural areas	Secondary
Fitness App Integrators	Developers and platforms consuming trek data via API	Tertiary

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Background and Motivation

2.1 Why This Product Is Needed

The global outdoor recreation market exceeds \$700 billion annually, with hiking and trekking as the fastest-growing segment. Smartphone penetration among trekkers exceeds 95%, yet no specialized application exists that accurately handles the unique technical challenges of mountain terrain. As health-conscious consumers demand richer fitness analytics, the inadequacy of repurposed road-navigation tools becomes increasingly apparent.

2.2 Existing Solutions & Their Limitations

Existing Solution	Key Limitation	Gap Addressed by TrekTrack AI
Google Maps / Waze	Designed for road navigation; 2D distance only; no trekking mode	3D terrain-aware distance + trail support
Strava	Activity misclassification in low-GPS zones; limited elevation analytics	AI-based activity classification + noise filtering
AllTrails	Static route database; no real-time adaptive tracking or AI	Real-time dynamic tracking with AI analytics
Garmin Connect	Requires dedicated hardware; not smartphone-native	Full-featured smartphone-only experience
Apple Health / Google Fit	Step-count only; no route or elevation intelligence	Full GPS route + elevation + terrain analytics

2.3 Market & Technical Gap

No existing mobile application combines: (a) 3D Euclidean distance calculation using real-time elevation data, (b) ML-based trekking activity classification resistant to GPS noise, and (c) comprehensive terrain analytics — all within a single offline-capable smartphone application targeted at trekkers.

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Goals and Objectives

3.1 Primary Goals

- Accurately calculate 3D trekking distance incorporating elevation gain/loss within $\pm 5\%$ of ground truth
- Correctly classify trekking vs. other movement types with $>95\%$ accuracy
- Provide real-time GPS noise filtering that reduces positional error to under 10 meters
- Deliver a mobile-first offline-capable experience with under 3-second UI response times

3.2 Secondary Goals

- Enable route sharing and community leaderboards for popular trekking regions
- Integrate with third-party health platforms (Apple Health, Google Fit, Strava)
- Support guided group trek sessions with live location sharing
- Provide educational terrain insights (slope difficulty ratings, altitude warnings)

3.3 Success Criteria

- MVP launched with core tracking features within 4 months of kickoff
- Beta user NPS score of 45 or above within 60 days post-launch
- 3D distance accuracy validated against professional GPS devices within 5% margin
- Activity classification accuracy $>95\%$ across a diverse labeled test dataset

3.4 Key Performance Indicators (KPIs)

KPI	Target	Measurement Method
3D Distance Accuracy	Within $\pm 5\%$ of reference GPS device	Field test comparison studies
Activity Classification Accuracy	$>95\%$ correct classification	Labeled dataset evaluation
GPS Noise Reduction	$<10\text{m}$ positional error (post-filter)	Static point deviation tests
App Crash Rate	$<0.1\%$ of sessions	Firebase Crashlytics
Session Start Time	<2 seconds to first GPS fix	Automated performance tests
User Retention (Day 30)	$>40\%$	Mixpanel cohort analysis
Daily Active Users (6-month)	50,000+ DAU	Analytics dashboard
App Store Rating	>4.5 stars	App Store / Play Store

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User Personas

4.1 Primary User — The Weekend Trekker

Persona: Arjun Sharma Age: 32 Software Engineer Mumbai, India
Motivation: Tracks weekend treks in the Sahyadri and Himalayan foothills. Wants accurate distance/elevation data to compare performance across treks.
Tech Comfort: High. Uses Strava and Google Fit but frustrated by misclassification and 2D-only distance.
Pain Points: Strava logs his 14 km mountain trek as 11 km (flat-map reading). Activity is sometimes classified as cycling. No offline maps for remote trails.
Needs From TrekTrack AI: Accurate 3D distance, offline trail maps, elevation profiles, and shareable trek summaries.

4.2 Primary User — The Trail Runner / Adventure Athlete

Persona: Priya Nair Age: 27 Physical Trainer Bangalore, India
Motivation: Competes in trail running events. Needs granular pace, elevation, and slope data for training optimization.
Tech Comfort: Very High. Uses multiple fitness apps simultaneously.
Pain Points: Existing apps lose GPS signal in forest sections. Elevation data from barometric altimeter on her watch conflicts with phone GPS readings.
Needs From TrekTrack AI: Real-time pace with slope-adjusted metrics, Kalman-filtered GPS, sensor fusion (barometer + GPS altitude).

4.3 Secondary User — The Trek Guide

Persona: Ravi Tomar Age: 45 Professional Trek Guide Manali, India
Motivation: Manages groups of 10-20 trekkers. Needs to monitor group progress and share routes with participants.
Tech Comfort: Moderate. Prefers simple UI but needs group coordination features.
Pain Points: No tool to broadcast real-time location to group members. Manual logging of route waypoints is time-consuming.
Needs From TrekTrack AI: Group session management, live location sharing (offline mesh or periodic sync), pre-loaded route broadcasts.

4.4 Secondary User — The Field Researcher

Persona: Dr. Meena Joshi Age: 39 Wildlife Biologist Pune, India
Motivation: Logs transect survey routes in national parks for biodiversity research. Needs exportable, timestamped GPS tracks.

Tech Comfort: Moderate. Values data accuracy over UI elegance.

Pain Points: Consumer apps don't support GPX/KML export. Location accuracy in dense forest is poor.

Needs From TrekTrack AI: GPX/KML export, raw data access, offline operation, configurable sampling rate.

05**User Stories****5.1 Core Tracking Stories**

1. As a weekend trekker, I want the app to automatically start recording my GPS path when I begin walking on a trail, so that I never miss the start of my trek.
2. As a trail runner, I want the app to calculate 3D distance including elevation changes, so that my reported distance reflects the true effort expended.
3. As a trekker, I want to see my live position on an offline map, so that I can navigate even without mobile data in remote areas.
4. As a trekker, I want the app to filter out GPS noise caused by tree cover, so that my recorded path does not show unrealistic zigzag patterns.

5.2 Analytics & Intelligence Stories

5. As a fitness-conscious trekker, I want to see a post-trek summary showing total distance, elevation gain, highest point, duration, and calories burned, so that I can evaluate my performance.
6. As a trail runner, I want slope-adjusted pace metrics (Grade-Adjusted Pace), so that I can compare efforts on steep vs. flat segments.
7. As a trekker, I want the app to detect when I'm trekking vs. traveling by vehicle, so that my activity log is correctly classified without manual intervention.
8. As a trekker, I want to view a 3D elevation profile of my completed route, so that I can understand the terrain I covered.

5.3 Social & Data Stories

9. As a trek guide, I want to create a group session and share my live location with all group members, so that no one gets lost during a guided trek.
10. As a researcher, I want to export my recorded trek as a GPX file, so that I can import it into GIS software for analysis.
11. As a trekker, I want to share a visual trek summary card on social media, so that I can celebrate my achievements with friends.
12. As a premium user, I want access to weather forecasts along my planned route, so that I can make informed decisions before starting a trek.

06**Functional Requirements**

6.1 Core Features

6.1.1 Real-Time GPS Tracking

- Continuous GPS coordinate recording at configurable intervals (1s, 5s, 10s)
- Background tracking with OS-level location service integration
- Altitude/elevation recording from GPS, barometric sensor, or DEM API (fallback hierarchy)
- Live polyline rendering on map canvas with position marker

6.1.2 3D Distance Calculation

- Euclidean 3D distance formula: $\sqrt{(\text{delta_lat})^2 + (\text{delta_lon})^2 + (\text{delta_alt})^2}$
- Cumulative distance aggregation updated in real time
- Separate tracking of horizontal distance and total 3D distance for comparison
- Segment-level distance breakdown for route analysis

6.1.3 Offline Map Support

- Pre-downloadable map tiles for user-defined geographic regions
- Support for OpenStreetMap, SRTM elevation data, and custom trail overlays
- Offline tile cache management with configurable storage limits

6.2 AI & Algorithmic Features

6.2.1 GPS Noise Filtering

- Kalman filter implementation for real-time positional smoothing
- Speed-based outlier rejection (remove points implying >50 km/h for trekking)
- Gaussian smoothing pass applied post-trek before analytics computation

6.2.2 Activity Classification

- ML classifier distinguishing: trekking, running, cycling, driving, stationary
- Input features: GPS speed, acceleration from IMU, elevation change rate, pace consistency
- Confidence score displayed to user; manual override always available
- Adaptive thresholds calibrated for mountainous GPS conditions

6.2.3 Terrain Analytics

- Real-time slope gradient calculation (percentage and degrees)
- Elevation profile generation with smoothed curve visualization
- Difficulty rating algorithm (Easy / Moderate / Hard / Expert) based on total gain, max slope, distance
- Grade-Adjusted Pace (GAP) calculation for runners

6.3 Data Processing Features

- GPX, KML, and JSON export of recorded tracks
- Post-trek smoothing and re-computation of stats after noise removal
- DEM (Digital Elevation Model) API integration for altitude correction
- Batch sync of offline-recorded sessions when connectivity is restored

6.4 User Interaction Features

- Trek start/pause/stop controls with haptic and visual feedback
- Waypoint marking with custom labels and photos
- In-trek stats overlay (distance, elevation, time, pace)
- Post-trek summary screen with shareable infographic generation
- Trek history browser with search, filter, and sort capabilities

6.5 Admin & Management Features

- Admin dashboard for monitoring app health, crash rates, and usage metrics
- Content moderation tools for shared routes and community posts
- User account management with subscription tier controls
- Feature flag system for staged rollout of new capabilities

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Non-Functional Requirements

Requirement	Specification	Priority
GPS Accuracy Post-Filter	< 10 meters positional error in open terrain	Critical
3D Distance Accuracy	Within ±5% of reference measurement	Critical
App Launch Time	< 2 seconds on mid-range devices	High
Background Battery Drain	< 8% per hour during active tracking	High
Offline Operation	Full tracking + map display without network	Critical
Data Sync Reliability	99.5% successful sync rate on reconnect	High
API Response Time	< 500ms for 95th percentile requests	High
Max Concurrent Users	100,000 simultaneous active sessions	High
Data Encryption	AES-256 at rest; TLS 1.3 in transit	Critical
GDPR/Privacy Compliance	Full compliance with data deletion rights	Critical
Accessibility	WCAG 2.1 AA compliant UI	Medium
Crash-Free Rate	> 99.9% sessions crash-free	High
Supported OS	iOS 14+ and Android 9.0+	Critical
Min Device Specs	2GB RAM, GPS chipset, 500MB storage	High

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System Architecture

8.1 High-Level Architecture Overview

TrekTrack AI follows a mobile-first, cloud-assisted architecture with robust offline-first design. The mobile application is the primary computing environment for all real-time tracking and AI inference. Cloud services provide synchronization, analytics aggregation, social features, and administrative functions.

8.2 Architecture Components

Layer	Component	Technology
Frontend (Mobile)	React Native cross-platform app	React Native 0.73+
Frontend (Mobile)	Map rendering engine	MapLibre GL / Mapbox SDK
Frontend (Mobile)	On-device ML inference	TensorFlow Lite / Core ML
Frontend (Mobile)	Local data store	SQLite + WatermelonDB
Backend API	RESTful + WebSocket API server	Node.js + Express / FastAPI
Backend API	Authentication service	Firebase Auth / Auth0
Backend API	Real-time group sessions	WebSocket (Socket.IO)
Database	User profiles & trek metadata	PostgreSQL (RDS)
Database	Time-series GPS tracks	TimescaleDB
Database	Map tile cache	Redis + CDN
AI/ML	Activity classifier training	Python + scikit-learn / TF
AI/ML	GPS filter (Kalman)	On-device (Dart/Swift/Kotlin)
External APIs	Elevation data	OpenTopoData / SRTM API
External APIs	Weather forecast	OpenWeatherMap API
External APIs	Third-party health sync	Apple HealthKit / Google Fit
Infrastructure	Cloud hosting	AWS (ECS, RDS, S3, CloudFront)
Infrastructure	CI/CD	GitHub Actions + Fastlane

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Data Requirements

9.1 Data Collected

Data Type	Fields	Source
GPS Track Points	latitude, longitude, altitude, accuracy, timestamp, speed	Device GPS
IMU Sensor Data	accelerometer x/y/z, gyroscope, barometer pressure	Device Sensors
User Profile	user_id, display_name, email, preferences, subscription_tier	User Input
Trek Session	session_id, start_time, end_time, activity_type, distance_2d, distance_3d	Computed
Elevation Model	DEM raster tiles, SRTM data at 30m resolution	OpenTopoData API
Map Tiles	Vector/raster tiles per geographic region	OSM / Mapbox
Analytics Events	screen_views, feature_usage, error_events	App Instrumentation

9.2 Data Storage Architecture

- On-device SQLite database stores active session data and offline map tiles — no data loss during connectivity outages
- Cloud PostgreSQL stores user profiles, trek metadata, and social data
- TimescaleDB (PostgreSQL extension) stores GPS time-series data with automatic partitioning by time
- S3 object storage for GPX/KML exports, generated infographics, and trek photos
- Redis caches elevation lookup results and user session tokens

9.3 Data Processing Pipeline

13. Raw GPS points streamed to on-device buffer (ring buffer, 1000-point capacity)
14. Kalman filter applied in real-time to each incoming point
15. Outlier detection removes physically impossible jumps (speed > 50 km/h)
16. 3D distance increments computed and accumulated per filtered point
17. On trek completion: Gaussian smoothing pass, DEM altitude correction, full stats recomputation
18. Processed track uploaded to cloud on connectivity restore; raw track retained locally for 30 days

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AI/ML Components

10.1 Activity Classification Model

Model Type

Gradient Boosted Trees (XGBoost) for initial deployment; LSTM-based sequence model for Phase 2 to capture temporal movement patterns.

Input Features

- GPS speed (km/h) — mean, std dev over 30s window
- Elevation change rate (m/min) — mean and variance
- Pace consistency score — coefficient of variation of inter-point intervals
- Accelerometer magnitude and variance from IMU
- GPS dilution of precision (DOP) score as terrain proxy

Output Classes

- TREKKING, TRAIL_RUNNING, CYCLING, DRIVING, STATIONARY, UNKNOWN

Training Data

- Minimum 10,000 labeled sessions across all activity classes
- Overrepresentation of mountainous and forested terrain recordings
- Data collected through beta user program with consent; supplemented by public GPS activity datasets (FitRec, Kaggle fitness datasets)

10.2 Kalman Filter (GPS Noise Reduction)

A standard linear Kalman filter with state vector [latitude, longitude, velocity_lat, velocity_lon] is implemented natively on-device. Process noise covariance is tuned for pedestrian movement. Measurement noise covariance is dynamically adjusted based on GPS accuracy radius reported by the device OS.

10.3 Elevation Correction

A hybrid elevation model combines: (1) device barometric altimeter for relative altitude changes (low noise, high precision), (2) GPS altitude as absolute reference, and (3) SRTM DEM as fallback and correction source. A Kalman-variant fusion filter blends barometer and GPS signals, corrected periodically against DEM lookups.

10.4 Model Evaluation Metrics

Model	Primary Metric	Target	Secondary Metric
Activity Classifier	Classification Accuracy	>95%	F1-score per class > 0.92
Activity Classifier	Confusion between TREKKING & DRIVING	<1%	Precision on TREKKING > 97%

Kalman GPS Filter	Mean Positional Error (static test)	<5m	90th percentile error < 10m
Elevation Fusion	Altitude RMSE vs. surveyed points	<8m	Drift over 10km route < 15m

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Technical Stack

Layer	Technology	Justification
Mobile Framework	React Native 0.73+	Single codebase for iOS & Android; large ecosystem; native module access
Map Rendering	MapLibre GL Native	Open-source; offline tile support; hardware-accelerated vector rendering
On-Device ML	TensorFlow Lite (Android) / Core ML (iOS)	Low-latency inference; no network required; model size <5MB
Local Database	WatermelonDB + SQLite	High-performance offline-first reactive data layer
Backend API	FastAPI (Python)	High performance async; ML-friendly; OpenAPI auto-docs
Auth	Firebase Authentication	OAuth 2.0, social login; proven scale; managed service
Primary DB	PostgreSQL 15 on AWS RDS	ACID compliance; PostGIS extension for geospatial queries
Time-Series DB	TimescaleDB	PostgreSQL-compatible; optimized for GPS time-series; auto-partitioning
Cache	Redis 7 on ElastiCache	Sub-millisecond elevation lookups; session token storage
Object Storage	AWS S3 + CloudFront CDN	GPX exports, images, map tile delivery at global edge
Real-Time	Socket.IO on Node.js	Group session live location broadcasting
ML Training	Python + scikit-learn + XGBoost + TensorFlow	Industry-standard ML pipeline; Jupyter notebooks for EDA
Cloud Platform	AWS (ECS, RDS, S3, CloudFront, CloudWatch)	Mature managed services; auto-scaling; global availability
CI/CD	GitHub Actions + Fastlane	Automated testing, building, and app store deployment
Monitoring	Datadog + Firebase Crashlytics + Sentry	Full-stack observability; real-time alerting
Analytics	Mixpanel	User behavior tracking; funnel analysis; cohort retention

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API Design

12.1 Core REST API Endpoints

Method + Endpoint	Description	Auth Required
POST /auth/register	Create new user account	No
POST /auth/login	Authenticate user, return JWT	No
GET /users/{id}/profile	Retrieve user profile and preferences	Yes
POST /sessions/start	Initialize a new trek session, return session_id	Yes
POST /sessions/{id}/points	Batch upload GPS points (offline sync)	Yes
POST /sessions/{id}/end	Finalize session, trigger server-side analytics	Yes
GET /sessions/{id}/summary	Get computed stats for a completed session	Yes
GET /sessions/{id}/track	Download full GPS track as GeoJSON	Yes
GET /sessions/{id}/export	Export track as GPX or KML file	Yes
GET /elevation/lookup	Get DEM elevation for lat/lng point or array	Yes
GET /users/{id}/history	Paginated list of user's past trek sessions	Yes
POST /groups/create	Create a guided group trek session	Yes
POST /groups/{id}/join	Join an existing group trek session	Yes
GET /groups/{id}/locations	WebSocket endpoint for real-time member locations	Yes

12.2 WebSocket Events

Event	Direction	Payload
location:update	Client -> Server	{ session_id, lat, lng, alt, timestamp }
location:broadcast	Server -> Clients	{ user_id, lat, lng, alt, timestamp }
session:status	Server -> Client	{ status: 'active' 'paused' 'ended' }
member:joined	Server -> Clients	{ user_id, display_name, avatar_url }

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UX/UI Requirements

13.1 Core Screens

Screen	Key Elements	Priority
Home / Dashboard	Recent treks, quick-start button, stats summary cards, weekly activity chart	Critical
Active Trek Map	Live map, position marker, polyline, stats overlay (distance, elevation, time), pause/stop FAB	Critical
Trek Summary	Route map thumbnail, 3D distance, elevation profile chart, pace breakdown, shareable infographic	Critical
Trek History	Chronological list with map thumbnails, search/filter bar, calendar view toggle	High
Trek Detail	Full route map, elevation chart, segment stats, export button, activity classification badge	High
Settings	GPS sampling rate, unit system (metric/imperial), offline map downloads, health app sync toggles	High
Group Session	Member list with live avatars on map, distance-to-group-center, join/create controls	Medium
Profile & Stats	Lifetime stats, trophies/badges, streak calendar, subscription management	Medium

13.2 Navigation Flow

- Bottom tab navigation: Home | Map | History | Groups | Profile
- Trek start initiates from Home (quick-start) or Map (full configuration flow)
- Active trek locks navigation to Map tab; other tabs accessible but with status bar
- Post-trek summary auto-shown on session end with share prompt

13.3 Design Principles

- Outdoor-Optimized: High-contrast color palette; readable in direct sunlight; large touch targets (min 48dp)
- Glanceable: Key metrics visible at a glance during active trek without unlocking phone
- Offline-First: All critical UI states function without network; clear 'offline' status indicators
- Progressive Disclosure: Simple defaults for casual users; advanced settings available but not surfaced by default
- Battery-Aware: Dark mode default to reduce OLED power consumption in field

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Risks and Challenges

Risk	Impact	Mitigation Strategy
GPS unavailability in deep valleys/canyons	High — tracking gaps, inaccurate distance	Dead reckoning using IMU; gap interpolation; user notification of signal quality
Activity classifier overfitting to training distribution	High — misclassification in new terrain types	Diverse training data; online learning from user corrections; conservative fallback to TREKKING
Excessive battery drain from continuous GPS	High — user abandonment	Adaptive sampling (reduce frequency when stationary); background mode optimization; battery usage screen
Altitude data inconsistency (GPS vs barometer vs DEM)	Medium — incorrect 3D distance	Sensor fusion with confidence weighting; DEM as ground truth validator; outlier rejection
Offline map storage exceeding device capacity	Medium — poor UX for low-storage devices	Progressive tile loading; storage limit warnings; selective region downloads
Privacy concerns with location data sharing in groups	High — regulatory risk, user trust	End-to-end encryption for group location; granular sharing controls; no storage of group tracks server-side without consent
App Store policy changes affecting background GPS	Medium — core feature breakage	Monitor Apple/Google OS release notes; maintain fallback motion-sensing mode
DEM/elevation API rate limits or downtime	Medium — elevation data unavailable	Local SRTM tile cache; multiple API provider fallbacks; graceful degradation to GPS-only altitude

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Development Roadmap

Phase 1 — MVP (Months 1–4)

Deliverable: Core tracking app with accurate 3D distance and offline maps
Week 1-2: Project setup, tech stack configuration, CI/CD pipeline, design system creation
Week 3-6: GPS tracking engine (Kalman filter + outlier rejection), 3D distance calculation
Week 7-8: Offline map integration (MapLibre + OSM tiles download manager)
Week 9-11: Trek session management (start/pause/resume/stop), on-device SQLite storage
Week 12-14: Post-trek summary screen, elevation profile chart, basic stats computation
Week 15-16: User auth (Firebase), cloud sync for completed sessions, beta release (TestFlight / Play Internal)

Phase 2 — Intelligence Layer (Months 5–8)

Deliverable: AI activity classification, terrain analytics, social sharing
Month 5: Activity classifier training pipeline, dataset collection from beta users, initial model (XGBoost)
Month 5-6: TFLite model deployment on-device, classification confidence UI, manual override flow
Month 6: DEM elevation correction API integration, barometer-GPS fusion algorithm
Month 6-7: Trek difficulty rating algorithm, Grade-Adjusted Pace for runners, segment breakdown
Month 7: Shareable trek summary infographic generator (canvas-based image export)
Month 7-8: GPX/KML export, Apple HealthKit + Google Fit integration, Strava webhook
Month 8: Trek history enhancements (search, filter, calendar view), public v1.0 app store launch

Phase 3 — Social & Growth (Months 9–12)

Deliverable: Group sessions, community features, premium tier, API
Month 9: Group trek session architecture, WebSocket real-time location sharing
Month 9-10: Route discovery feed (community-shared treks), region-based leaderboards
Month 10: Premium subscription tier (advanced analytics, unlimited offline maps, weather forecast)
Month 10-11: LSTM-based sequence model upgrade for activity classification
Month 11: Public developer API + documentation (trek data consumption by third parties)
Month 12: International expansion: multi-language support, region-specific trail databases

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Future Enhancements

Enhancement	Description	Timeline
Augmented Reality Navigation	AR trail waypoint overlays using phone camera for navigation at forks and summits	Year 2
Wearable Integration	Full sync with Apple Watch, Wear OS, Garmin; heart rate + HRV analytics	Year 2
AI Route Recommendation	Personalized trail suggestions based on fitness level, location, and past performance	Year 2
Safety SOS System	One-tap emergency alert with last-known GPS location sent to emergency contacts and services	Year 1.5
Weather Micro-Forecast	Hyperlocal weather warnings (storm, lightning) based on real-time route and elevation	Year 1.5
Trek NFTs / Digital Badges	Blockchain-verified achievement badges for summit completions and milestones	Year 3
Offline AI Voice Assistant	On-device voice assistant for hands-free trek queries and navigation commands	Year 3
B2B Guide Platform	White-label version for commercial trekking operators with booking and fleet management	Year 2
Carbon Footprint Tracker	Calculate and offset the carbon footprint of travel to/from trek starting points	Year 2

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Metrics for Success

17.1 Product Quality Metrics

Metric	Definition	Target (Year 1)
3D Distance Accuracy	Absolute % error vs. reference GPS device in field tests	< 5%
Activity Classification Accuracy	% correctly classified sessions in test dataset	> 95%
GPS Smoothing Effectiveness	Mean positional error post-filter in static tests	< 10m
Crash-Free Rate	% of app sessions that complete without a crash	> 99.9%
GPS Fix Acquisition Time	Seconds from app open to first GPS lock	< 5s (95th pct)

17.2 User Engagement Metrics

Metric	Definition	Target (Year 1)
Monthly Active Users	Unique users completing at least 1 trek per month	100,000
Day-30 Retention	% of Day-0 installs still active on Day 30	> 40%
Sessions per Active User	Average trek sessions per user per month	> 3
Net Promoter Score	User satisfaction and recommendation likelihood	> 50
App Store Rating	Average store rating on iOS + Android	> 4.5 stars

17.3 Business Metrics

Metric	Definition	Target (Year 1)
Total Downloads	Cumulative installs across iOS + Android	500,000
Premium Conversion Rate	% of MAU converted to paid premium subscription	> 5%
MRR Growth (Monthly)	Month-over-month recurring revenue growth rate	> 15% MoM
GPX Export Volume	Number of GPX files exported per month (proxy for power users)	> 10,000/mo
Group Session Adoption	% of MAU creating or joining group sessions	> 8%

TrekTrack AI — PRD v1.0

This document is intended for internal use by engineering, design, and product teams.

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